1 SNPE-C | Reference obs. 1 | dim(θ) = 4

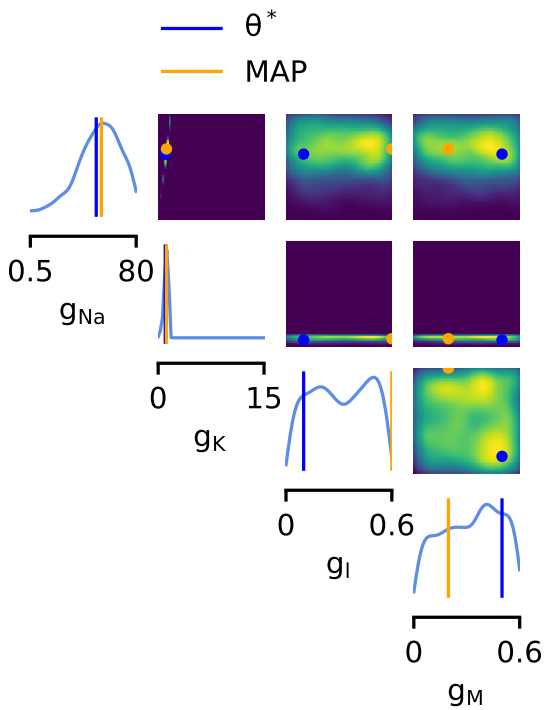

Figure 1: Posterior distribution over Hodgkin-Huxley parameters g_{Na}, g_K, g_l, g_M.

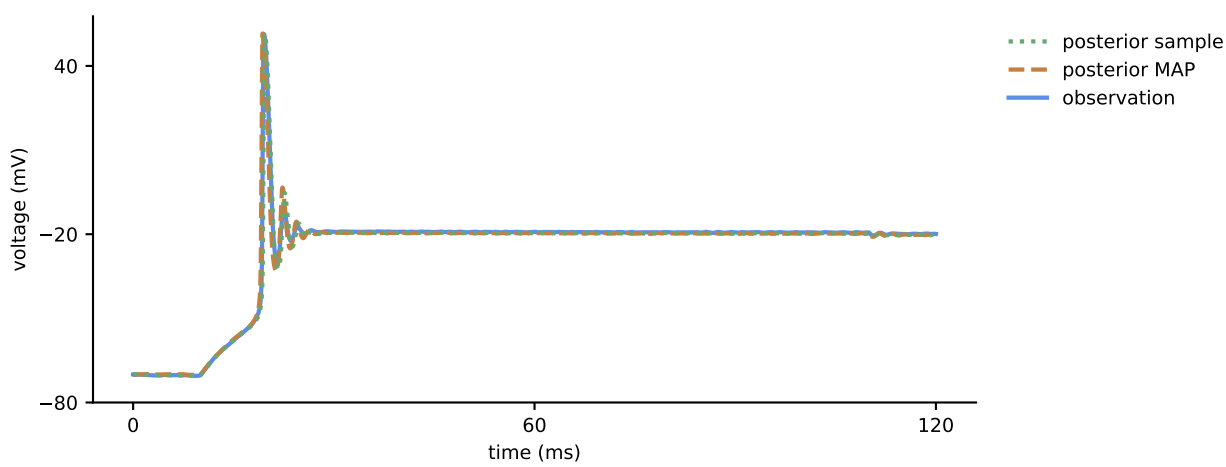

Figure 2: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

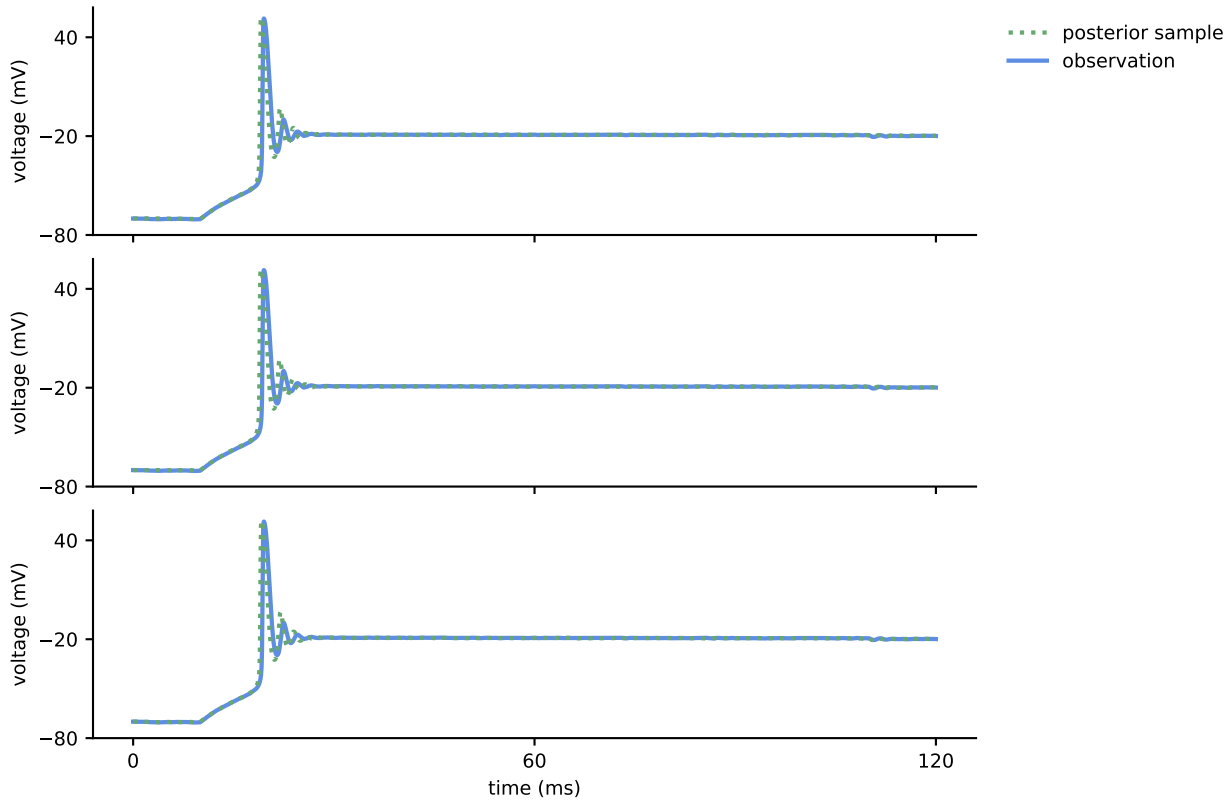

Figure 3: Simulation outputs from three parameters drawn from the posterior.

2 ESNPE | Reference obs. 1 | dim(θ) = 4

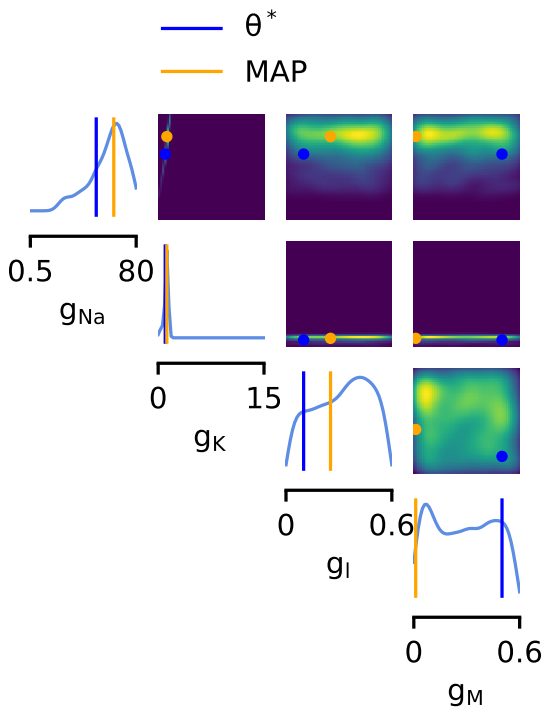

Figure 4: Posterior distribution over Hodgkin-Huxley parameters g_{Na}, g_K, g_l, g_M.

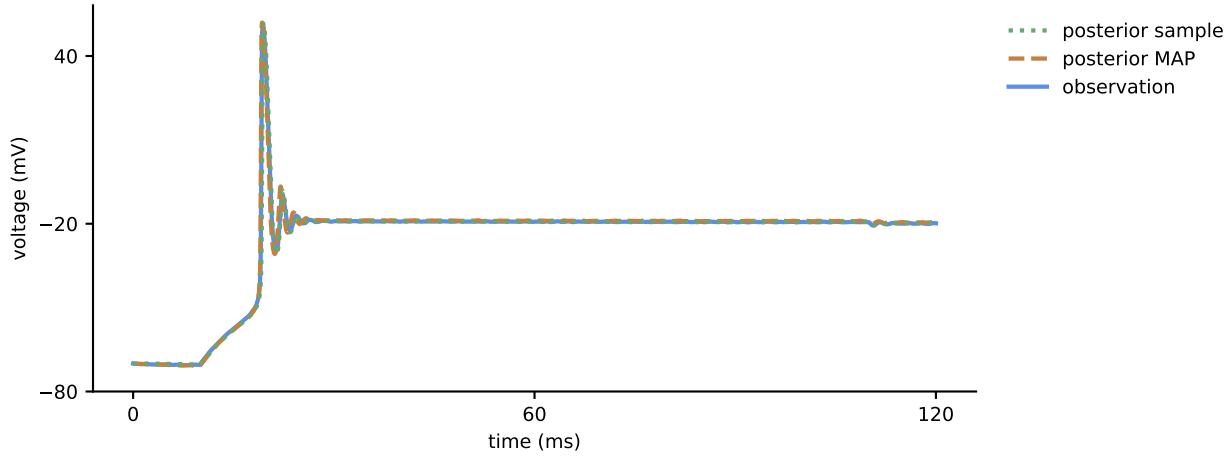

Figure 5: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

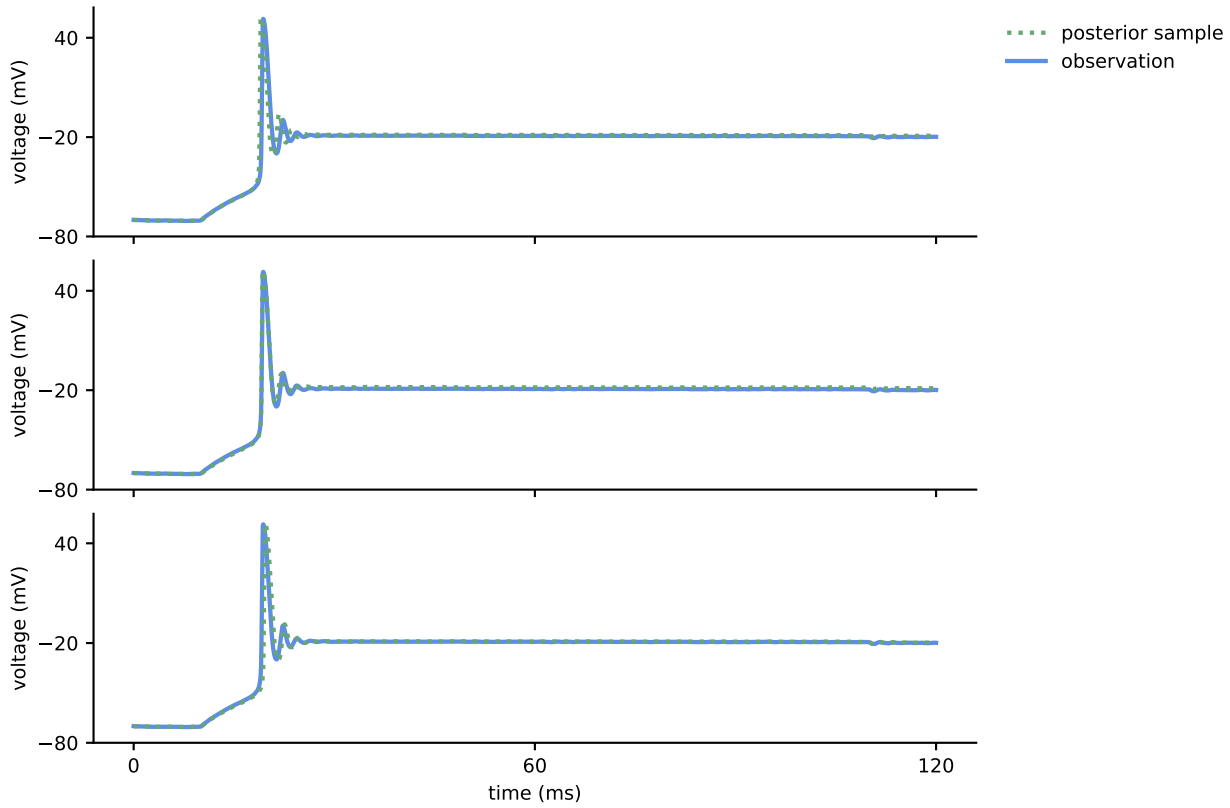

Figure 6: Simulation outputs from three parameters drawn from the posterior.

3 SNPE-C | Reference obs. 2 | $\dim(\theta) = 4$

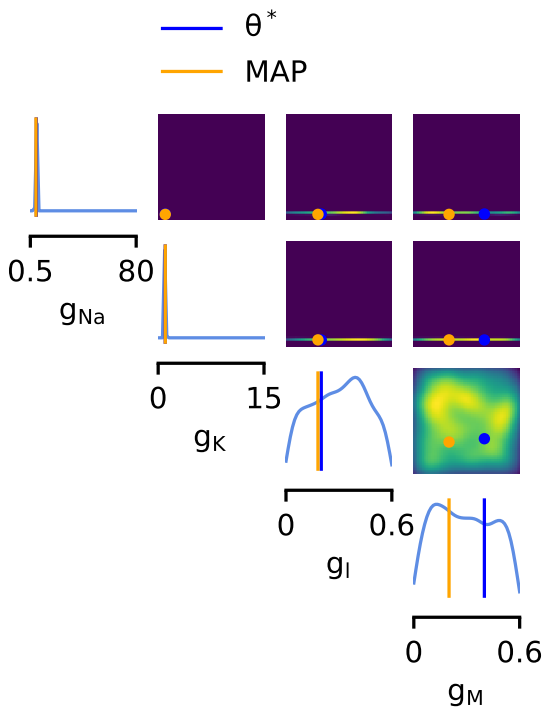

Figure 7: Posterior distribution over Hodgkin-Huxley parameters g_{Na}, g_K, g_l, g_M.

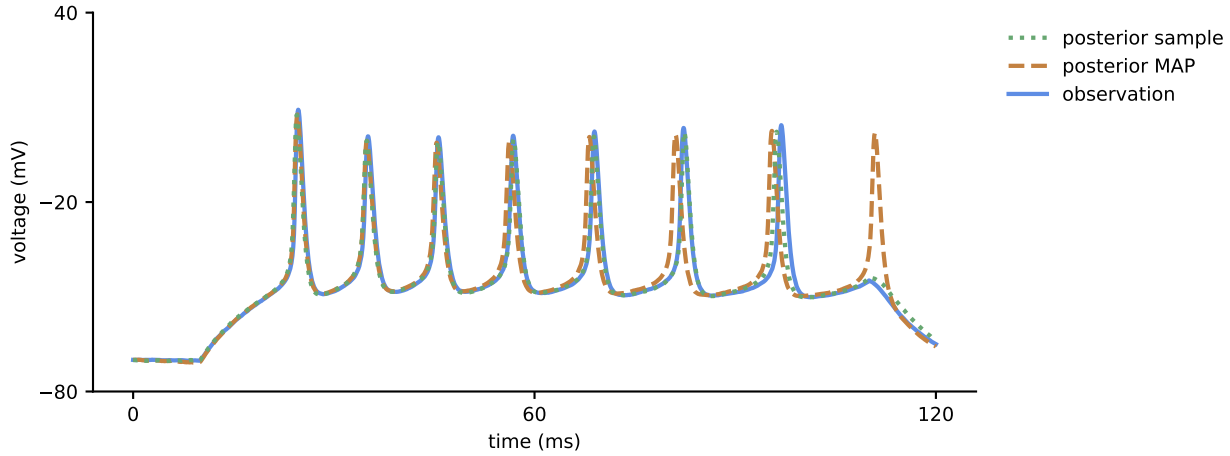

Figure 8: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

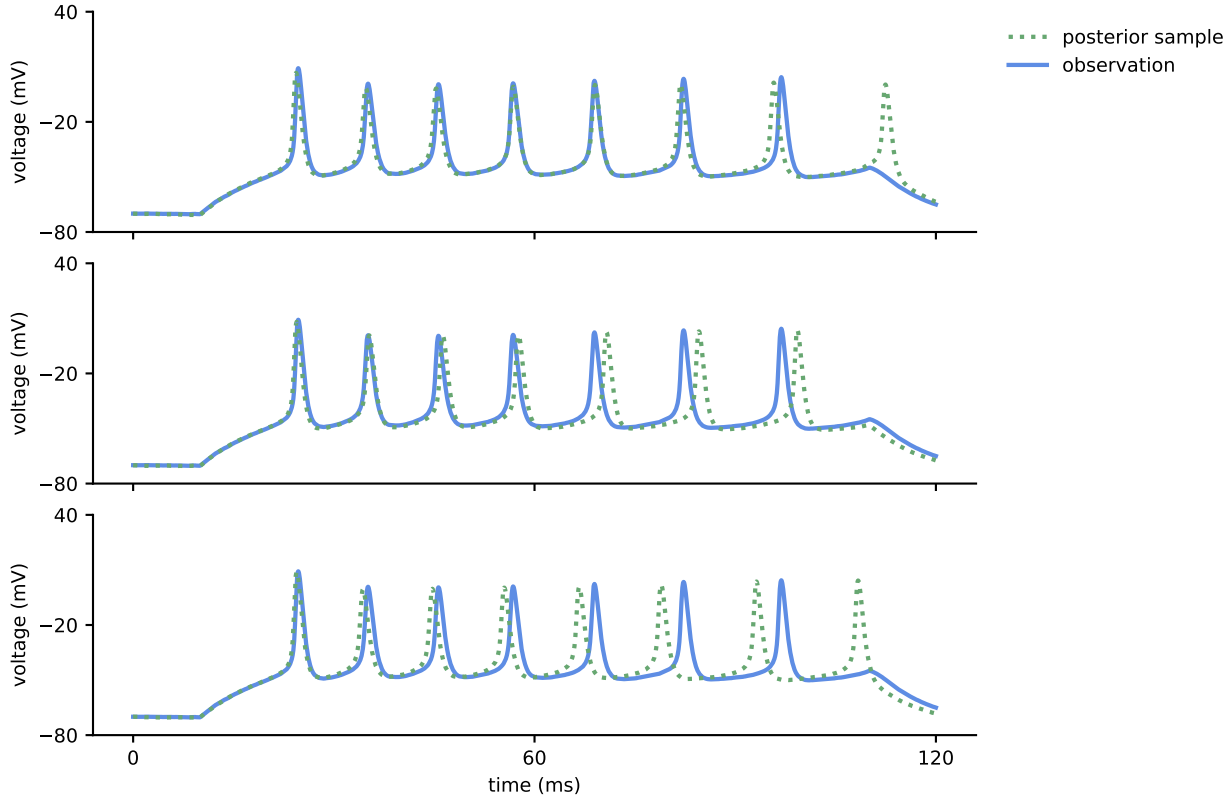

Figure 9: Simulation outputs from three parameters drawn from the posterior.

4 ESNPE | Reference obs. 2 | dim(θ) = 4

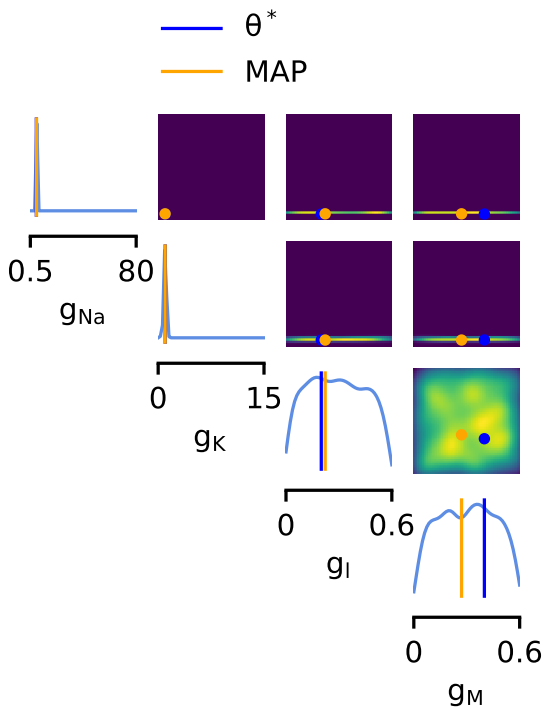

Figure 10: Posterior distribution over Hodgkin-Huxley parameters g_{Na}, g_K, g_l, g_M.

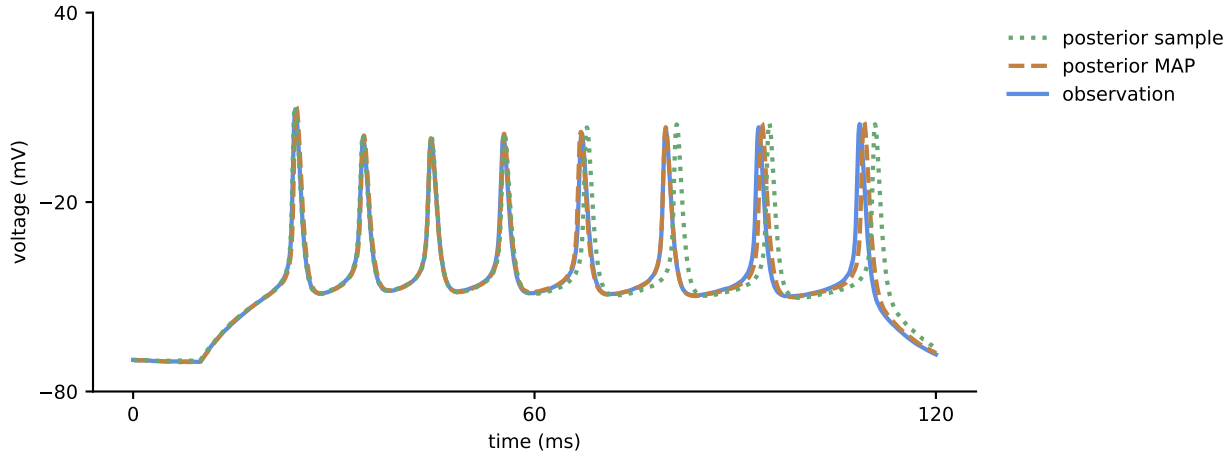

Figure 11: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

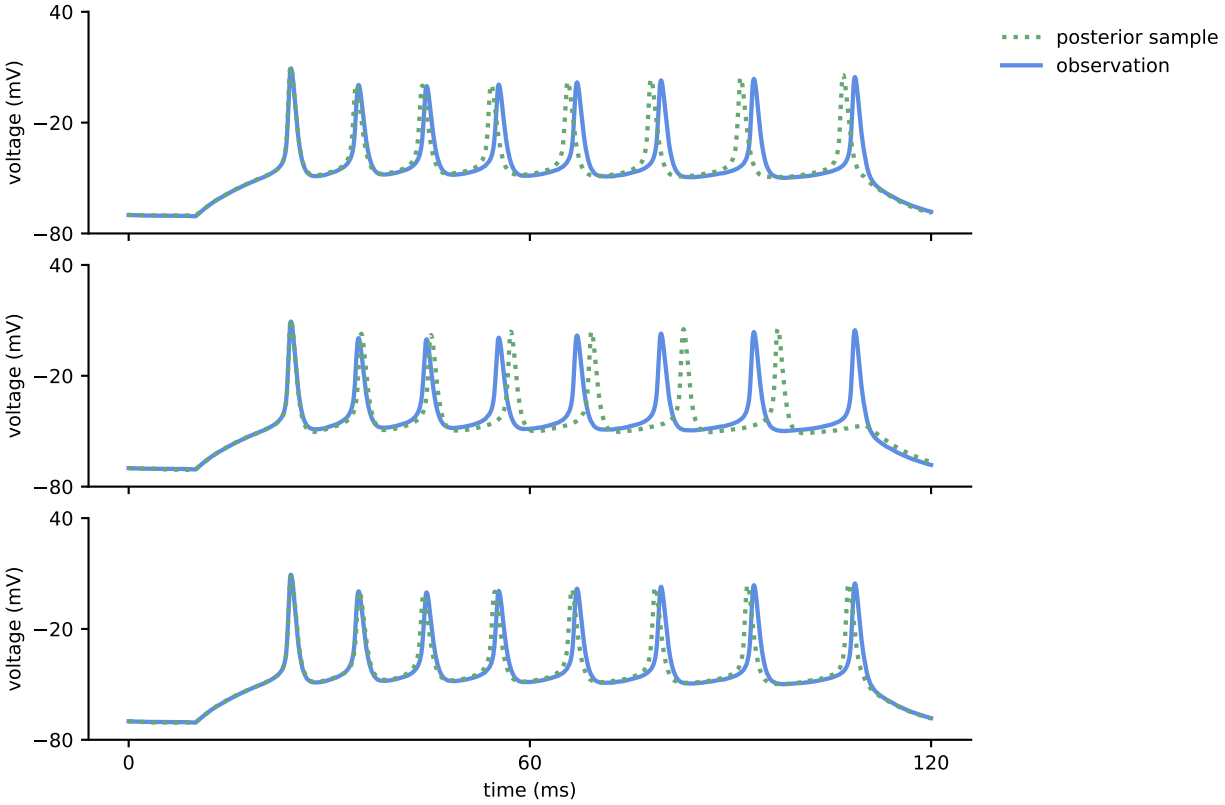

Figure 12: Simulation outputs from three parameters drawn from the posterior.

5 SNPE-C | Reference obs. 1 | dim(θ) = 8

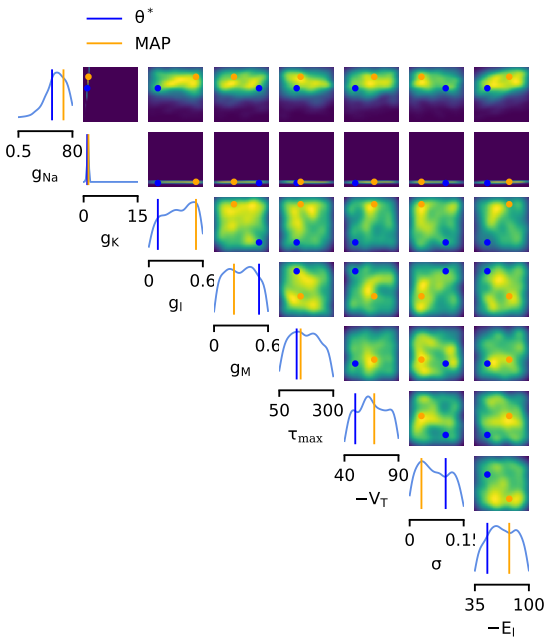

Figure 13: Posterior distribution over Hodgkin-Huxley parameters $g_{Na}, g_K, g_l, g_M, \tau_{\max}, -V_T, \sigma, -E_l$.

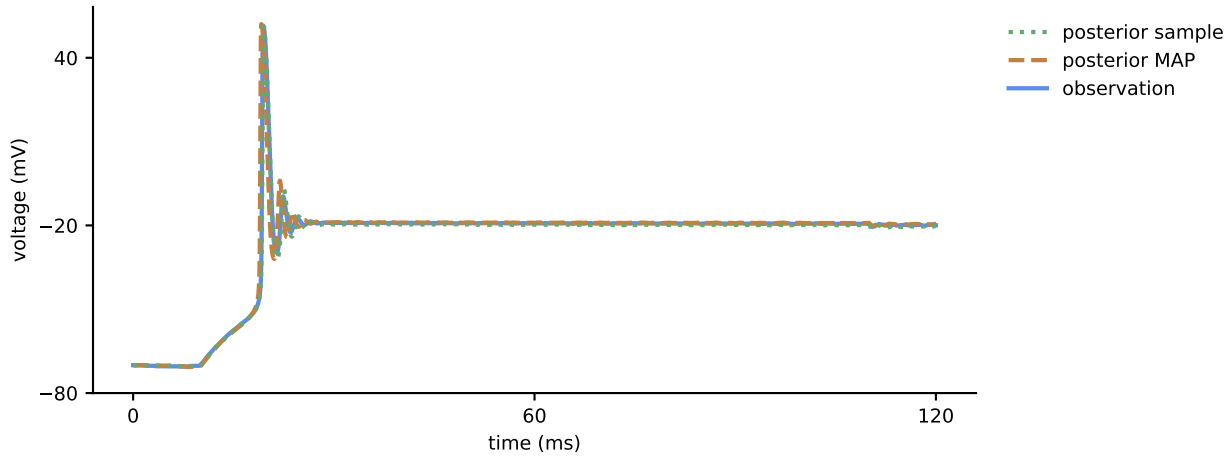

Figure 14: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

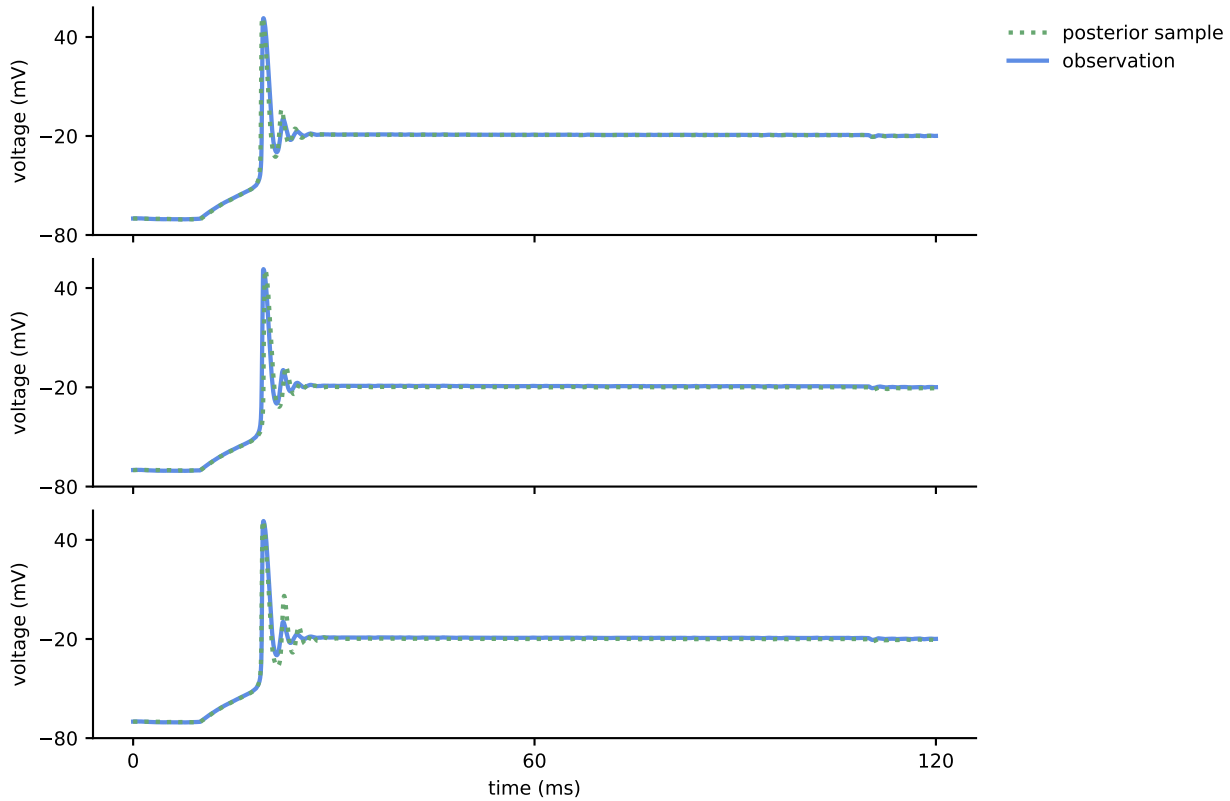

Figure 15: Simulation outputs from three parameters drawn from the posterior.

6 ESNPE | Reference obs. 1 | dim(θ) = 8

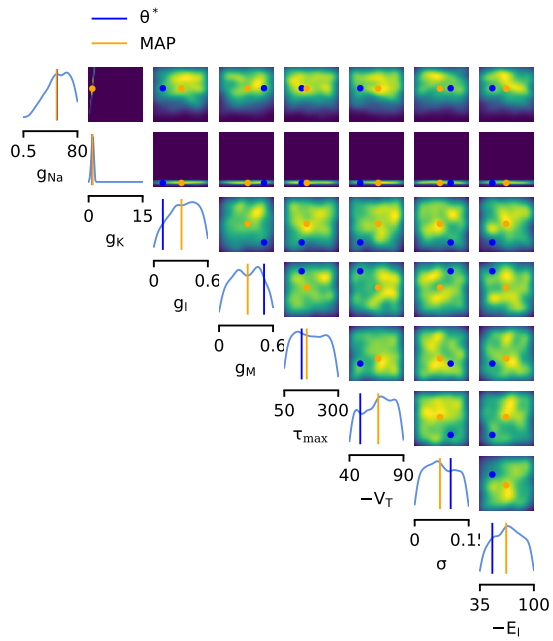

Figure 16: Posterior distribution over Hodgkin-Huxley parameters $g_{Na}, g_K, g_l, g_M, \tau_{\max}, -V_T, \sigma, -E_l$.

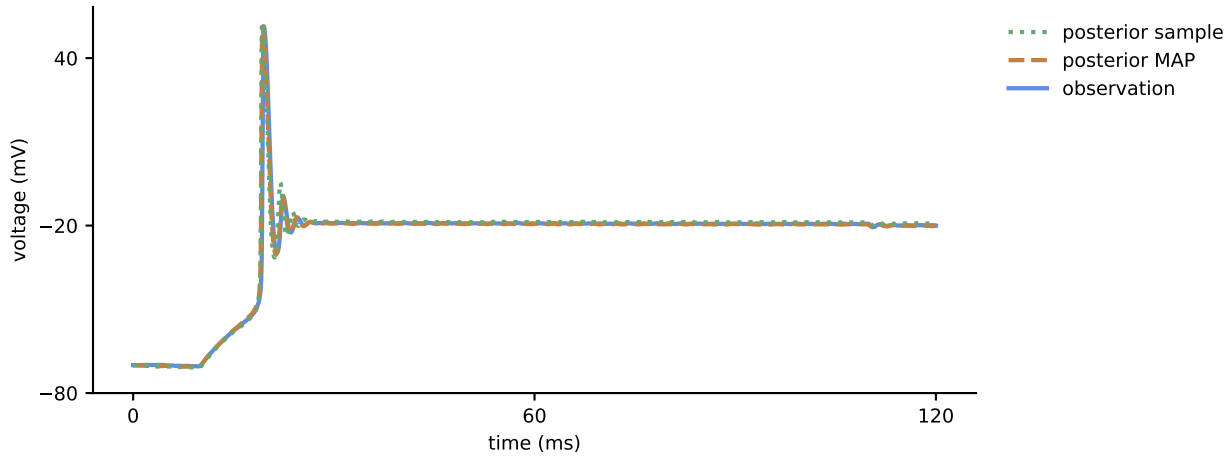

Figure 17: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

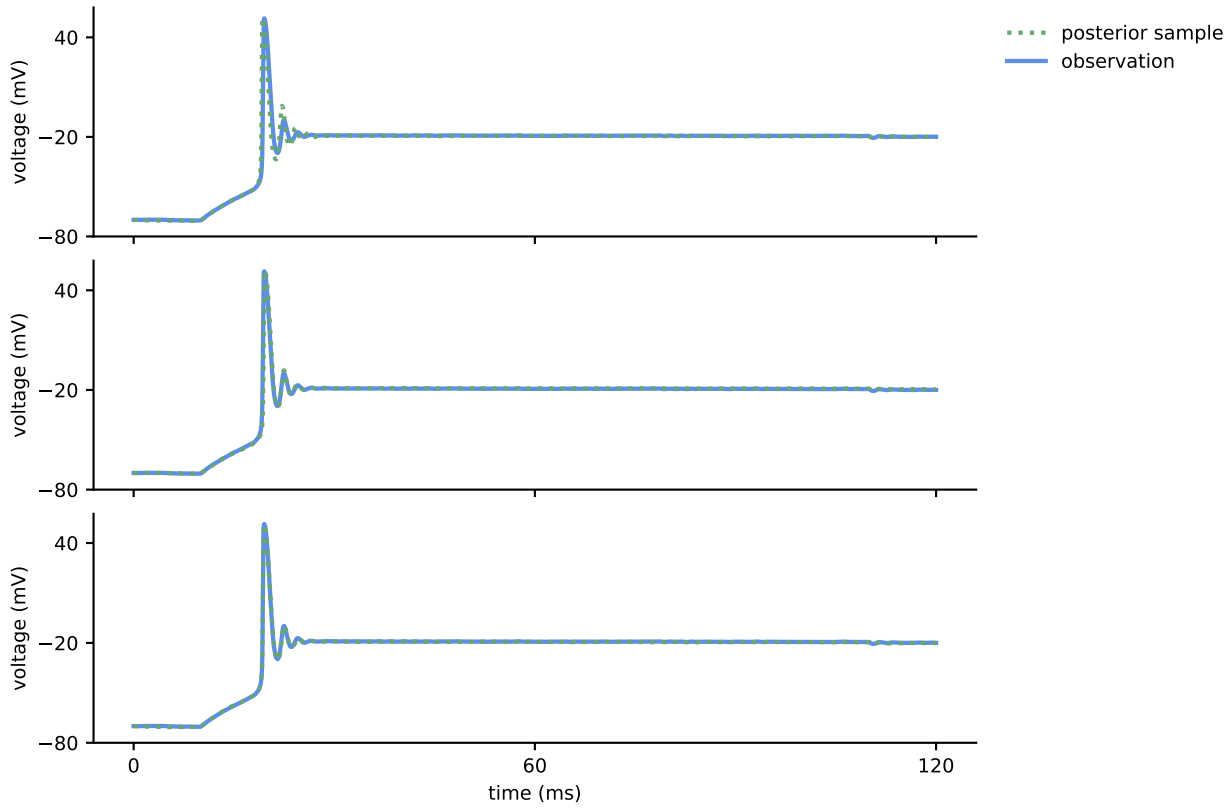

Figure 18: Simulation outputs from three parameters drawn from the posterior.

7 SNPE-C | Reference obs. 2 | dim(θ) = 8

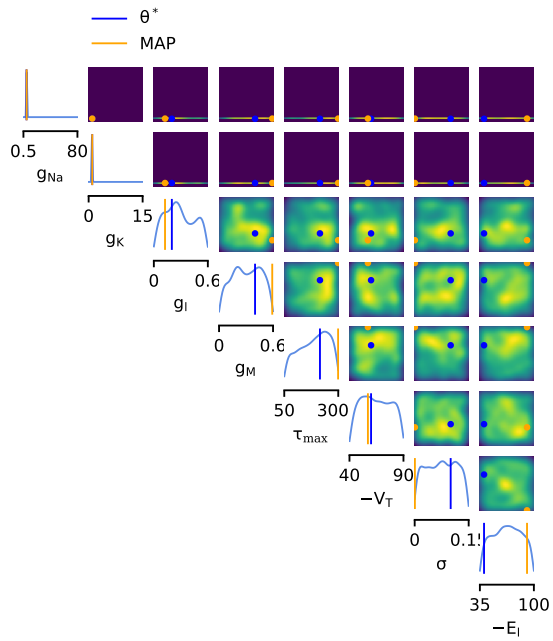

Figure 19: Posterior distribution over Hodgkin-Huxley parameters $g_{Na}, g_K, g_l, g_M, \tau_{\max}, -V_T, \sigma, -E_l$.

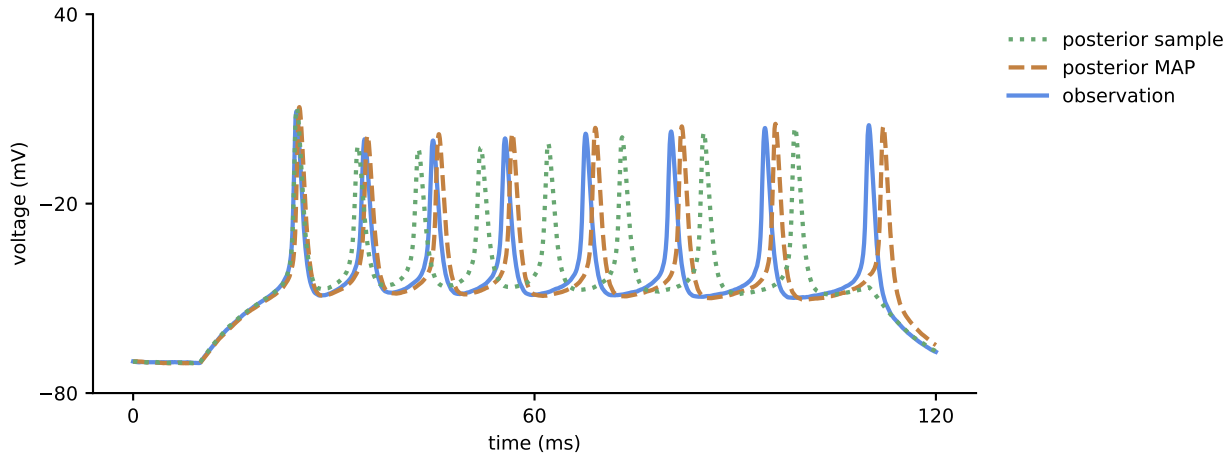

Figure 20: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

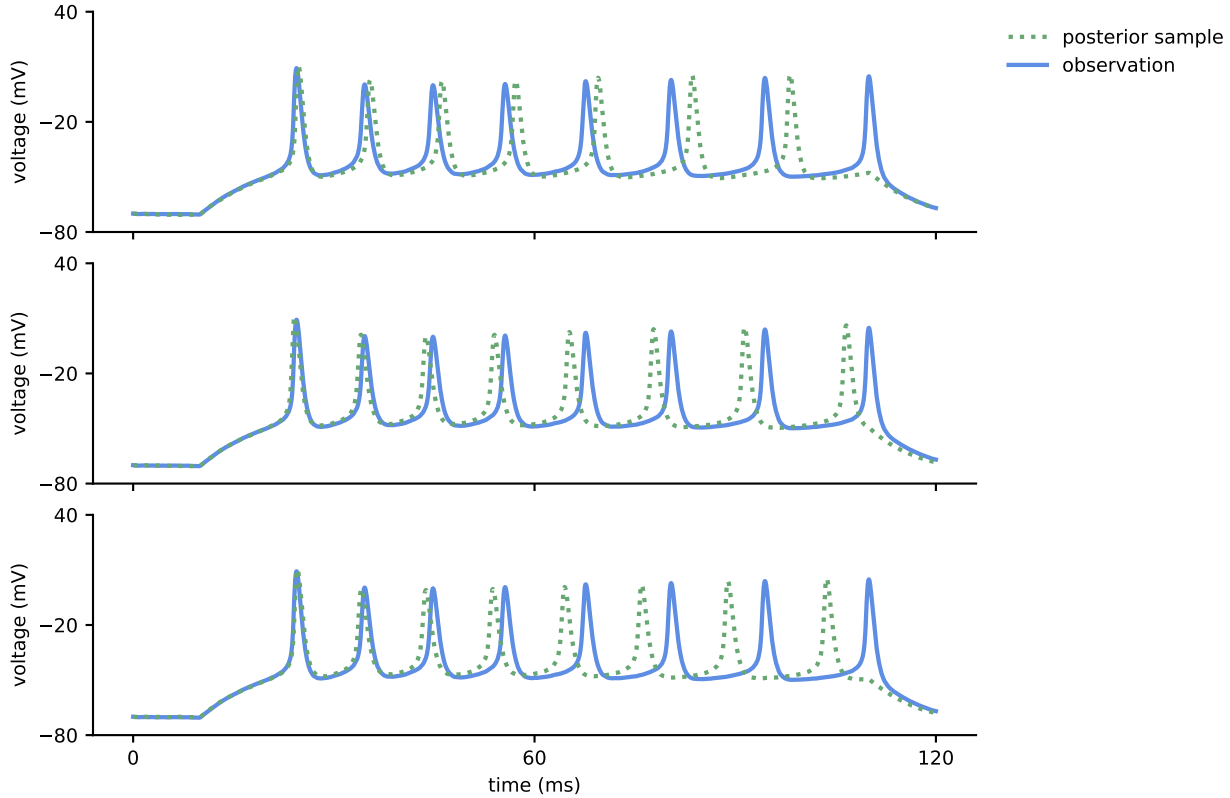

Figure 21: Simulation outputs from three parameters drawn from the posterior.

8 ESNPE | Reference obs. 2 | dim(θ) = 8

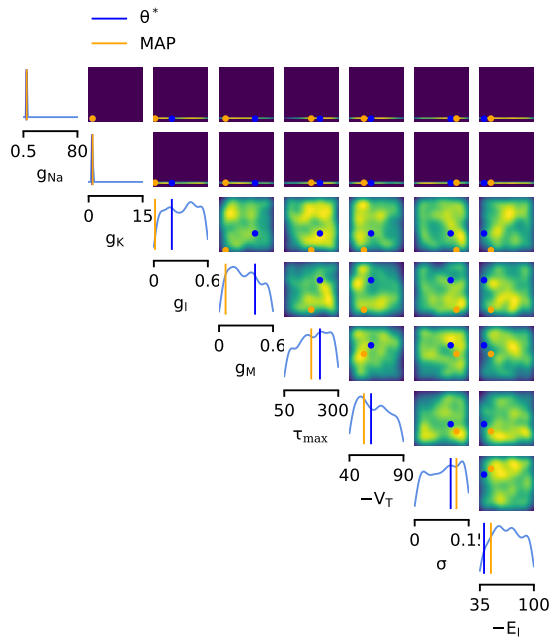

Figure 22: Posterior distribution over Hodgkin-Huxley parameters $g_{Na}, g_K, g_l, g_M, \tau_{\max}, -V_T, \sigma, -E_l$.

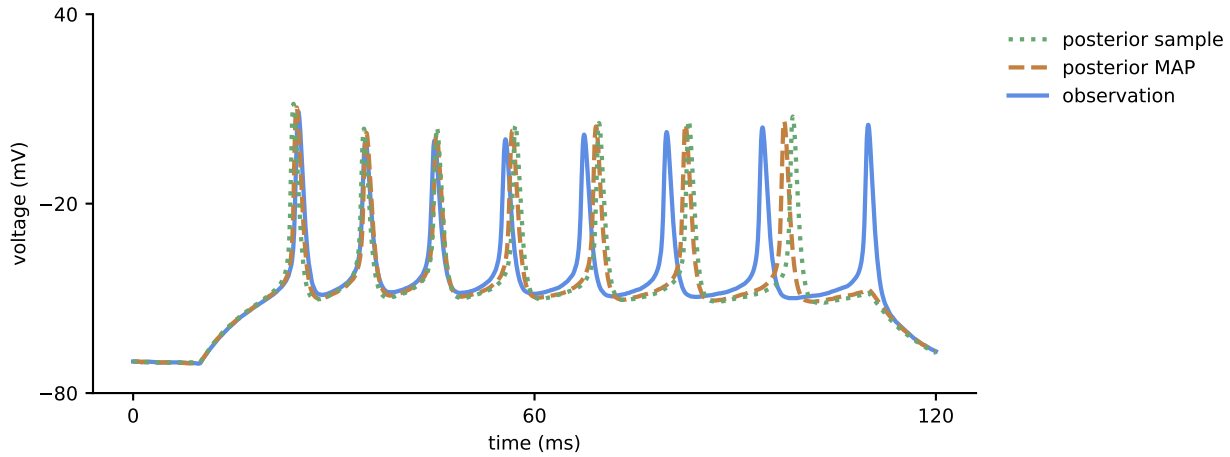

Figure 23: Simulator output comparison drawn under the reference parameter θ^* (blue, solid), approximate posterior MAP (orange, dashed), and a randomly drawn parameter from the posterior (green, dotted).

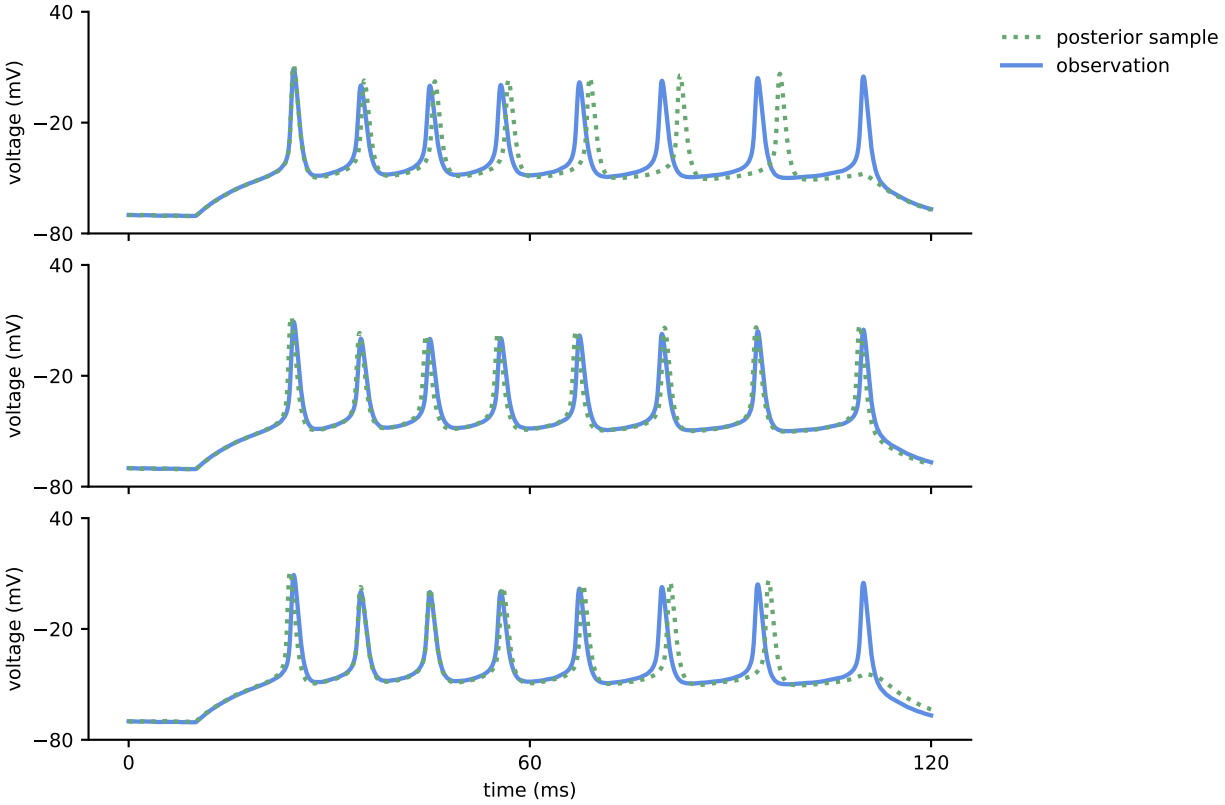

Figure 24: Simulation outputs from three parameters drawn from the posterior.